

Adaptive Bi-Recommendation and Self-Improving Network for Heterogeneous Domain Adaptation-Assisted IoT Intrusion Detection

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Abstract—As Internet of Things (IoT) devices become prevalent, using intrusion detection to protect IoT from malicious intrusions is of vital importance. However, the data scarcity of IoT hinders the effectiveness of traditional intrusion detection methods. To tackle this issue, in this article, we propose the adaptive bi-recommendation and self-improving network (ABRSI) based on unsupervised heterogeneous domain adaptation (HDA). The ABRSI transfers enrich intrusion knowledge from a data-rich network intrusion source domain to facilitate effective intrusion detection for data-scarce IoT target domains. The ABRSI achieves fine-grained intrusion knowledge transfer via adaptive bi-recommendation matching. Matching the bi-recommendation interests of two recommender systems (RSs) and the alignment of intrusion categories in the shared feature space form a mutual-benefit loop. Besides, the ABRSI uses a self-improving mechanism, autonomously improving the intrusion knowledge transfer from four ways. A hard pseudo label (PL) voting mechanism jointly considers RS decision and label relationship information to promote more accurate hard PL assignment. To promote diversity and target data participation during intrusion knowledge transfer, target instances failing to be assigned with a hard PL will be assigned with a probabilistic soft PL, forming a hybrid pseudo-labeling strategy. Meanwhile, the ABRSI also makes soft pseudo-labels globally diverse and individually certain. Finally, an error knowledge learning mechanism is utilized to adversarially exploit factors that causes detection ambiguity

and learns through both current and previous error knowledge, preventing error knowledge forgetfulness. Holistically, these mechanisms form the ABRSI model that boosts IoT intrusion detection accuracy via HDA-assisted intrusion knowledge transfer. Comprehensive experiments on several intrusion data sets demonstrate the state-of-the-art performance of the ABRSI method, outperforming its counterparts by 9.2%, and also verify the effectiveness of ABRSI constituting components and ABRSI's overall efficiency.

Index Terms—Adaptive bi-recommendation (ABR), domain adaptation (DA), Internet of Things (IoT), intrusion detection, self-improving (SI).

I. INTRODUCTION

AS Internet of Things (IoT) devices evolve rapidly, they transform various applications to become more intelligent [1], [2], [3]. However, IoT tends to be computational and energy constrained, which prevents some powerful intrusion detection mechanisms from being deployed, together with a lack of regular maintenance, these limitations make IoT devices vulnerable to intrusions [4]. Hence, an effective intrusion detection mechanism is necessary to protect IoT infrastructures from malicious intrusions.

Previous research efforts attempted to tackle the IoT intrusion detection (IID) problem mainly via the rule-based and machine learning-based perspectives. For instance, Jun and Chi [5] and Dietz et al. [6] proposed to use complex event processing and a proactive violation scanning mechanism to detect events that match violation patterns defined in a rule repository. On the other hand, intrusion detectors based on ML models, such as stacked autoencoder [7], multikernel SVM [8], and capsule network [9], etc., have been proposed and achieved satisfying performance. However, these efforts either require a sophisticated rule repository which is highly expertise-dependent and can hardly be thorough, or depend on a fully annotated training data set, which is labor-intensive to build. Moreover, the limited storage and communication capability and the involvement of user privacy make IoT data hard to be obtained and publicly released. Together with the cost of expertise and manual annotation, IoT tends to be data

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scarce [10], hindering the usefulness of rule and ML-based methods under IoT scenarios.

Recently, domain adaptation-based (DA) intrusion detection methods are proposed, which transfer intrusion knowledge from the data-rich network intrusion (NI) domain to facilitate more effective IoT intrusion (II) detection under data scarcity. By aligning both domains into a common feature space, the enriched intrusion knowledge from the source NI domain can benefit the target II intrusion detection. Besides, a heterogeneous DA-based (HDA) method [11] should also be able to tackle interdomain heterogeneities, such as different feature representations, different feature dimensions, etc. For example, Hu et al. [12] proposed a deep subdomain adaptation network which transferred intrusion knowledge via local maximum mean discrepancy minimization. Xie et al. [13] presented a collaborative alignment framework (CAF), which reduced global domain discrepancy and preserves local semantic consistency. Liang et al. [14] put forward the Pareto DA (PDA) method that controls the overall optimization direction for better intrusion knowledge transfer.

Despite the effectiveness of these HDA-based methods, they still have some limitations that need to be addressed.

- 1) The recommender system (RS) is an effective way to exploit “interest,” i.e., intrusion knowledge of intrusion categories. However, to the best of our knowledge, RS and bi-recommendation matching haven’t been introduced to improve HDA-based intrusion detection.
- 2) The pseudo label (PL) can exploit the potential of unlabeled target II domain. However, to our best knowledge, none of the previous method assigned PL based on the knowledge gained from RS due to their complete ignorance of the RS. Besides, previous methods assigned PL in an isolated manner without considering label relationship information.
- 3) When utilizing PL, previous methods either use it in a one-hot way, i.e., hard PL [15], or assign a probabilistic soft PL [14], [16]. These two ways either hurt participation and diversity of target II instances during knowledge transfer, or lacked enough emphasis of confident and correct PL. None of them considered a hybrid PL strategy that jointly involves both hard and soft PL for better participation, diversity and enough emphasis of confident and correct predictions.
- 4) Finally, although some methods performed intrusion knowledge transfer via minimizing the divergence of intrusion category probabilistic correlation semantics [17], i.e., error knowledge, they didn’t learn it from an adversarial manner and simultaneously consider both the current and past error knowledge, resulting in error knowledge forgetfulness.

To address these limitations and enhance the intrusion knowledge transfer, by following the high-level intuition illustrated in Fig. 1, we propose an adaptive bi-recommendation and self-improving (ABRSI) network in an unsupervised HDA setting, i.e., the target II domain is completely unsupervised, meanwhile heterogeneities are present between source NI and target II domains. To mine the intrusion category interest and benefit intrusion knowledge transfer, we utilize an adaptive

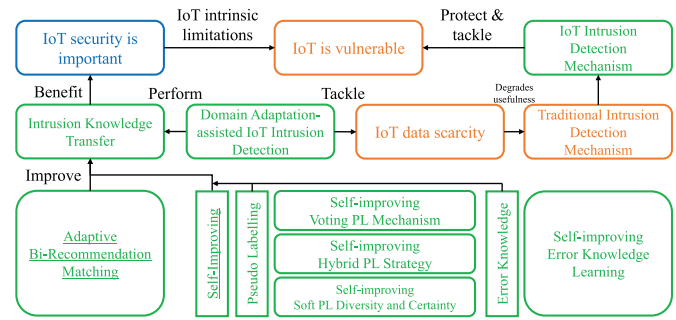


Fig. 1. Intuition of the ABRSI model.

bi-recommendation (ABR) matching mechanism. The ABR leverages two RSs, one for each domain, and matches the recommendation for each intrusion category between two RSs, which enforces feature alignment in the shared feature space. Better feature alignment in turn can produce enhanced RS. By adaptively iterating this learning process, it forms a mutually beneficial positive loop. Then, the ABRSI method also involves self-improving (SI), which autonomously improves the intrusion knowledge transfer in four ways. First, we jointly consider the neural network prediction, RS decision and label relationship information to improve PL accuracy and prevent negative transfer caused by error-prone PLs. A hard PL will only be assigned if these votes reach a consensus, promoting confident and correct PL assignment as much as possible. Second, the hard-PL-only strategy prevents unlabeled target instances from participating the intrusion knowledge transfer if a hard PL is not assignable. The diversity of hard PL may also be impaired as hard PLs tend to focus on easy intrusion categories. Thus, we utilize a hybrid PL strategy that involves both hard and probabilistic soft PL to emphasize confident and correct predictions, and meanwhile improve participation and diversity. Third, we use the Tsallis entropy and a diversity maximization mechanism, aiming for globally diverse and individually certain soft PL. Finally, the ABRSI applies an error knowledge learning (EKL) mechanism, which emphasizes error knowledge, i.e., factors that cause prediction ambiguities, and tackling it in an adversarial manner. Meanwhile, both current and previous error knowledge are involved to prevent error knowledge forgetfulness. Holistically, it forms the ABRSI framework that can effectively transfer intrusion knowledge for better intrusion detection.

In summary, the contributions of this article are fourfold.

- 1) We realize the usefulness to transfer enriched intrusion knowledge from source NI domain to facilitate better intrusion detection for target II domain under an extreme data-scarce unsupervised HDA setting.
- 2) To our best knowledge, we are the pioneer to introduce a RS to facilitate better intrusion knowledge transfer. An ABR matching mechanism is used to match interests of intrusion categories, forming a positive loop in which bi-recommendation matching and feature space alignment can mutually benefit each other adaptively. The RS also benefits hard PL assignment by avoiding PLs that violate mined intrusion interests.

- 3) We utilize a SI mechanism with four submechanisms, i.e., SI hard PL accuracy via voting, SI participation and diversity via hybrid PL strategy, SI soft PL diversity, and confidence and SI by exploiting and eliminating error knowledge, and meanwhile prevent error knowledge forgetfulness.
- 4) We conduct comprehensive experiments on five widely recognized intrusion detection data sets to verify the effectiveness of ABRSI, and meanwhile show the usefulness of each constituting component and verify the efficiency.

The remainder of this article is organized as follows. Section II summarizes related works by categories and present our research opportunities, followed by Section III, which presents model preliminaries and the ABRSI architecture. Section IV presents the detailed mechanism constituting the ABRSI model. Experimental setups and result analyses are given in Section V. Section VI concludes this article. We provide an acronym table and a notation table for better readability.

II. RELATED WORK

Traditional IID: IID has drawn wide attention from the research community. The rule-based methods perform intrusion detection based on a predefined intrusion rule repository. For instance, Jun and Chi [5] proposed complex event processing, which can reactively match events with predefined intrusion rules. Later, Dietz et al. [6] proposed to detect intrusions proactively by periodically scanning for predefined malicious behaviors among connected IoT devices. As ML techniques evolve quickly, they have been widely adopted to perform intrusion detection. Possible models including isolation forest [18], multikernel SVM [8], and deep learning methods, such as capsule network [9], stacked autoencoder [7], etc. However, these methods either depend on a thorough and up-to-date rule repository, which requires sophisticated expertise to construct with a high cost, or depend on a fully labeled training data set, which is time and labor-intensive to annotate, and is, especially, difficult to acquire due to IoT data scarcity caused by insufficient resource capabilities and associated data privacy issues. Therefore, it provides rooms for DA-based solutions, which can tackle the data-scarce IoT scenario.

HDA: HDA-based methods can transfer knowledge from a knowledge-rich source domain to a knowledge-scarce target domain, given that heterogeneities are present between domains. Hence, it is applicable to facilitate more effective intrusion detection under data-scarce IoT scenarios. Vu et al. [19] proposed an autoencoder-based method, which aligned the bottleneck layer to achieve intrusion knowledge transfer. Hu et al. [12] presented a deep subdomain adaptation network and transferred intrusion knowledge via local maximum mean discrepancy minimization. There are some other HDA methods, although not specifically targeting the intrusion detection scenario, their excellent knowledge transfer capability can be utilized to tackle the intrusion detection problem and achieve excellent performance. Xie et al. [13]

proposed a CAF that performed global domain alignment via Wasserstein distance minimization (CAFD) or adversarial learning (CAFA). It also achieved knowledge transfer via minimizing the distance between probabilistic outputs yielded by two classifiers, a way to learn from errors. Li et al. [17] considered the JADA model, which jointly achieved both global alignment and local error minimization between probabilistic outputs of two differently initialized classifiers. However, neither the JADA nor CAF method performed probabilistic output matching via a previous knowledge-enabled adversarial way, and their lack of PL information makes their matching process coarse-grained. Li et al. presented DCAN [20] and GDCAN [21], which transferred knowledge via domain conditional channel attention and the adaptive channel attention mechanism, respectively. However, these methods lacked enough focus on PL assignment, making their knowledge transfer become coarse-grained. On the other hand, Long et al. [15] proposed conditional domain adversarial network (CDAN) to transfer knowledge via a conditional generative adversarial network. It also utilized hard PL assignment generated by the classifier network. Jin et al. [16] put forward a minimum class confusion (MCC) loss as a pluggable tool to generalize and improve the knowledge transfer, assisted by a classifier-predicted hard PL strategy. Lv et al. [14] achieved knowledge transfer via PDA, a Pareto optimal solution searching method assisted by soft PL assignment. However, these aforementioned methods either assign hard PL in an isolated manner without considering label relationship information, or fail to consider the hybrid PL strategy, resulting in compromised target instance participation and diversity, or a lack of emphasis on confident and correct predictions. Recently, Li et al. [22] presented the progressive alignment framework, which alleviated feature discrepancy via shared codebook learning, and meanwhile minimized the distribution divergence in a progressive manner. This unified framework achieved superior knowledge transfer performance. Li et al. [23] further considered a more generalized scenario, i.e., the divergence-agnostic scenario in which either source or target domain data is unknown. They then proposed the AAA model to tackle this scenario via adversarial attacks. However, all of these methods completely ignored the benefit brought by the RS.

RS and Intrusion Detection: To our best knowledge, we are the pioneer to introduce the RS to enhance the intrusion knowledge transfer. To find a comparable baseline, we adapt the transfer learning-based RS algorithm [24] and form ADAR, which utilizes domain discriminator to fuse the RS feature matrix, then perform intrusion detection via the trained RS. However, the ADAR does not involve a recommendation matching mechanism, and fails to utilize the RS to produce more accurate PLs.

Research Opportunity: The ABRSI method jointly utilizes the ABR mechanism and a SI mechanism. The ABR mechanism forms a mutually beneficial positive loop between intrusion feature alignment and interest matching of two RSs. The SI mechanism benefits fine-grained intrusion knowledge transfer from four ways: it prevents error-prone hard PL with compromised diversity by improving the PL accuracy through a voting mechanism, the RS also contributes during PL voting.

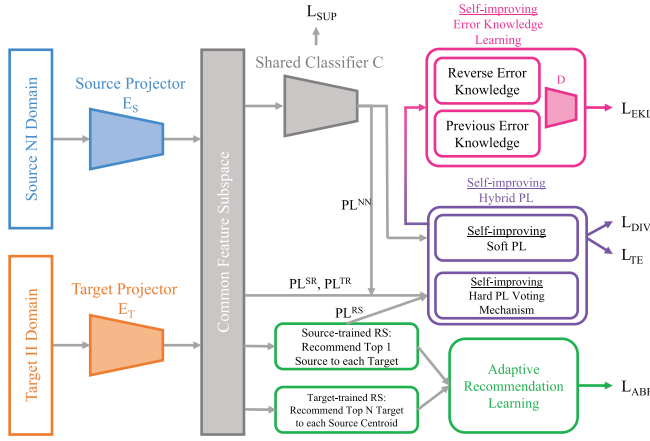


Fig. 2. Architecture of the ABRSI model.

Hence, it performs better than the isolated PL strategy used in previous methods; it improves participation and diversity via a hybrid PL strategy, which fills the void of previous methods; it improves the diversity and certainty of soft PL; it improves EKL and jointly considers previous error knowledge to prevent error forgetfulness, which is lacked by previous methods.

III. MODEL PRELIMINARY AND ARCHITECTURE

A. Model Preliminary

The ABRSI model works under the unsupervised HDA setting. We follow common notations [10], [25] to define the source NI domain $\mathcal{D}_S$ as follows:

$$\mathcal{D}_S = \{\mathcal{X}_S, \mathcal{Y}_S\} = \{(x_{S_i}, y_{S_i})\}, i \in [1, n_S]$$

$$x_{S_i} \in \mathbb{R}^{d_S}, y_{S_i} \in [1, K] \quad (1)$$

where $\mathcal{X}_S$ is the set of source NI traffic data, $\mathcal{Y}_S$ is the corresponding intrusion category label, n_S and d_S denote the number of source NI instances and dimension of source NI domain, respectively. K denotes the total number of intrusion categories. Similarly, the target II domain has the following definition:

$$\mathcal{D}_T = \{\mathcal{X}_T\} = \{(x_{T_i})\}, i \in [1, n_T], x_{T_i} \in \mathbb{R}^{d_T}. \quad (2)$$

The ABRSI model works under the unsupervised HDA setting, i.e., the target II domain is completely unlabeled. Besides, heterogeneities present between domains, e.g., $d_S \neq d_T$.

B. ABRSI Architecture

The architecture of the ABRSI model is illustrated in Fig. 2. Each domain has a feature projector that maps heterogeneous features into a d_C -dimensional shared feature subspace. The feature projector E has the following definition:

$$f(x_i) = \begin{cases} E_S(x_i), & \text{if } x_i \in \mathcal{X}_S \\ E_T(x_i), & \text{if } x_i \in \mathcal{X}_T \end{cases}$$

$$f(x_i) \in \mathbb{R}^{d_C}. \quad (3)$$

A shared classifier C is used to make intrusion detection decisions. The ABRSI utilizes the ABR mechanism. It involves a source-trained RS that recommends the top one source

instance to each target, and a target-trained RS that recommends the top N target instances to each source category centroid. Matching the categorywise recommendation between two RSs can enforce finer feature alignment in the shared feature space, and it in turn yields RS with an improved recommendation capability. Hence, it forms a positive loop in which bi-recommendation matching and feature alignment mutually benefit each other in an adaptive way. The ABRSI then leverages the SI mechanism from four perspectives. A pseudo-labeling voting mechanism is used to improve the accuracy of hard PL assignment by jointly considering the classifier prediction, RS decision and label relationship information. It accounts for the intrusion interests learned by the RS and avoids error-prone near boundary PLs benefitted from label relationship information. To enhance label diversity and promote better participation of unlabeled target instances during intrusion knowledge transfer, the ABRSI adopts a hybrid PL strategy. The Tsallis entropy and diversity maximization loss is also used to make the soft PL globally diverse and individually confident. Finally, the ABRSI uses an EKL mechanism, which exploits error knowledge that causes intrusion detection ambiguities and iteratively eliminates it via adversarially learning with reverse error knowledge and previous error knowledge. The EKL holistically achieves error knowledge exploitation, elimination, and meanwhile prevents error forgetfulness. Forming these mechanisms into a holistic model achieves fine-grained intrusion knowledge transfer so that the shared classifier C produces the best intrusion detection efficacy for the unlabeled target II domain.

IV. ABRSI ALGORITHM

A. Adaptive Bi-Recommendation Matching

RS is a powerful tool to exploit interests, i.e., characteristics of each intrusion category. However, the RS for each domain cannot find similar interests without the features being well-aligned as illustrated in Fig. 3 (left). The ABR bridges the gap between bi-recommendation matching and fine-grained intrusion domain alignment. The ABRSI utilizes a source-trained RS, denoted as RS_S . Since both domains are mapped into a common feature space, therefore, the source-trained RS should be able to recommend the most similar source intrusion instance to a given target instance. If the RS fully exploits intrusion category interests, then, recommending an intrusion type B source instance to a given unlabeled target instance indicates that target instance is highly likely to belong to intrusion type B, i.e., the top one recommendation for each unlabeled target instance forms a RS-based PL PL_{RS} . Note that in this case, only the top one recommendation is used since it needs to act as the PL for unlabeled target instances. By averaging target instances based on its RS-based PL, it forms the intrusion categorywise RS recommendation for the target domain. Similarly, a target-trained RS_T is utilized to recommend top N most similar target instances to each source category centroid, and take the average of these N recommendations as a final recommendation for each source intrusion category. Note that in this case, the top N recommended target instances are used for a better representativeness of target

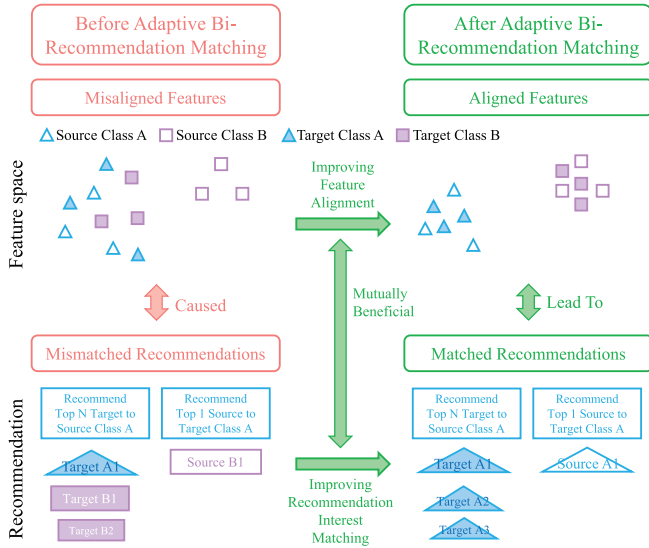


Fig. 3. ABR matching and feature alignment mutually benefit each other, forming a positive loop that enhances fine-grained intrusion knowledge transfer.

domain interests. By enforcing the categorywise recommendation to be similar between two RSs, it in turn forces the intrusion domains to align in the shared feature space, as illustrated in Fig. 3 since misaligned feature space can only cause mismatched RS recommendations. Moreover, the aligned features can also facilitate the bi-RS to better exploit and match the intrusion category interests, hence, it forms a positive loop in which bi-recommendation interest matching and feature alignment mutually benefit each other as the training progresses. As shown in Fig. 3 (right), the ABR mechanism will eventually achieve a well-aligned recommendation interest matching and a fine-grained feature space alignment.

The ABRSI model utilizes latent semantic indexing (LSI) algorithm as its RS. Unlike neural network-based RS algorithms that involve a tremendous amount of parameters [26], the LSI is not heavily data-hungry during training unlike its neural-network-based counterparts, and enjoys a low computational complexity during intrusion detection inference. Besides, it has been leveraged in various recommendation system applications [27], [28] and achieved a satisfying recommendation effectiveness, which makes it suitable to be used under data-scarce and computationally constrained IoT scenarios. The formulation of LSI is as follows:

$$M_S \approx U_S^R T_S^R V_S^{R\top}, U_S^R = \begin{bmatrix} x_S^{1'} \\ x_S^{2'} \\ \vdots \\ x_S^{n'} \end{bmatrix}, x_T^j = x_T^j U_S^R T_S^{R\top}$$

$$M_S \in \mathbb{R}^{d_C \times n_S}, U_S^R \in \mathbb{R}^{d_C \times R}, T_S^R \in \mathbb{R}^{R \times R}, V_S^{R\top} \in \mathbb{R}^{n_S \times R}$$

$$x_S^i, x_T^j \in \mathbb{R}^{d_C}, x_S^{i'}, x_T^{j'} \in \mathbb{R}^R \quad (4)$$

where M_S is the original feature matrix of the source NI domain, R is the reduced dimension parameter, U_S^R , T_S^R , and $V_S^{R\top}$ are the feature-latent, latent transfer, and instance-latent matrix for the source domain D_S , respectively. $x_S^{i'}$ denotes

the transformed i th source NI instance. Then, the top one recommendation for each target II instance x_T^j is made as follows:

$$RS_S(x_T^j) = x_S^i, PL_{x_T^j}^{RS} = y_S^i, i = \operatorname{argmax}_i \cos(x_S^{i'}, x_T^{j'}). \quad (5)$$

The target-trained RS_T is defined similarly with the above definition of RS_S . The RS_S will recommend the most similar source instance $RS_S(x_T^j)$ to each target instance x_T^j , its category label acts as the RS label $PL_{x_T^j}^{RS}$, and the RS_T will recommend the top N most similar target instances $RS_T(x_S^i)^n, n \in [1, N]$ to each source category centroid x_S^i . The ABR will then minimize the divergence between categorywise bi-recommendations as follows:

$$\mu_{RS_S}^k = \operatorname{mean}(RS_S(x_T^j)), \text{ where } PL_{x_T^j}^{RS} = k$$

$$\mu_{RS_T}^k = \sum_{n=1}^N \frac{(RS_T(x_S^i)^n)}{N}, \text{ where } y_S^i = k$$

$$\mathcal{L}_{ABR} = \sum_{k=1}^K \frac{\|\mu_{RS_S}^k - \mu_{RS_T}^k\|_2^2}{K} \quad (6)$$

where $\mu_{RS_S}^k$ denotes the centroid of target instances that are recommended with a category k source instance, and $\mu_{RS_T}^k$ denotes the centroid of target instances recommended to the centroid of source category k . As the training progresses, the recommendations produced by bi-RSs will become more reliable, and, hence, the ABR matching is gradually emphasized by a linearly growing hyperparameter ρ . The ABR loss $\mathcal{L}_{ABR}$ minimizes the categorywise divergence between bi-recommendations, which, therefore, promotes better feature space alignment, forming a positive loop as indicated in Fig. 3.

B. Self-Improving Mechanism

Hard PL Voting: To give unlabeled target instances a chance to participate during fine-grained intrusion knowledge transfer, assigning hard PLs to them is a common way to fully exploit their potentials. However, previous methods tend to directly use the error-prone network predicted PL, which misleads the intrusion knowledge transfer and causes negative transfer. To mitigate their drawbacks, the ABRSI uses a hard PL voting mechanism. First, the ABRSI will use the RS decision $PL_{x_T^j}^{RS}$ introduced earlier to avoid confident but wrong network predicted PLs that violates the interests mined by the RS, as indicated in Fig. 4 (b). We are the pioneer to use a RS to improve hard PL accuracy for better intrusion knowledge transfer. Then, the ABRSI will consider the PL relationship with surrounding source instances. If the neighboring source instances cannot reach a consensus, or the consensus contradicts with the network predicted PL or the RS decided PL, then that PL is highly likely to reside near the decision boundary, i.e., ambiguous near-boundary PL, as indicated in Fig. 4(c). Hence, the ABRSI will avoid this assignment when the source label relationship information yields disagreements. Finally, the ABRSI performs unsupervised clustering on unlabeled target instances to reveal the target label relationship. If a target instance has a PL assignment that contradicts with

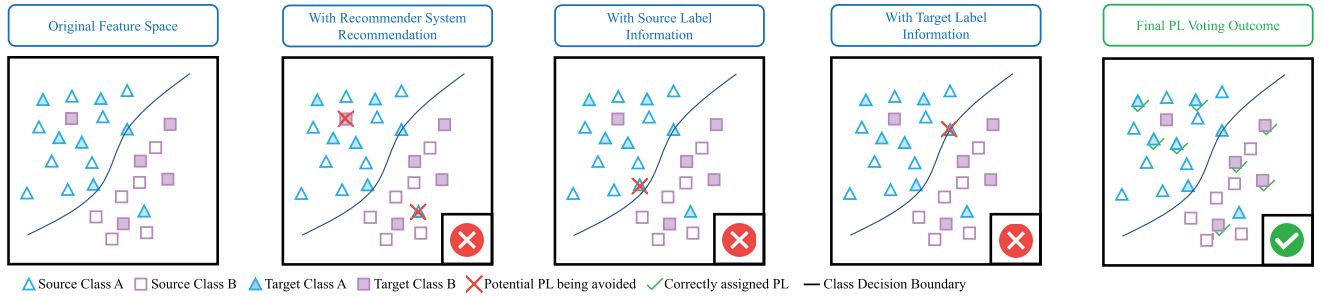


Fig. 4. PL voting mechanism. (a) Original feature space. (b–d) PLs eliminated by RS recommendation, source and target label relationship information, respectively. (e) Hard PL assigned by PL voting mechanism.

TABLE I
HYBRID PL STRATEGY AND ITS ADVANTAGES OVER HARD AND SOFT-PL-ONLY STRATEGIES

Strategy	Use Hard PL	Use Soft PL	Pros & Cons
Hard PL only	✓	✗	✓ Emphasis on confident and correct PLs ✗ Error prone and misleading ✗ Instances without hard PL cannot participate during alignment ✗ Hard PLs favour easy instances, lacking diversity
Soft PL only	✗	✓	✓ All instances can participate during alignment ✓ Promotes better diversity using probabilistic output as PL ✗ Confident and correct predictions get less emphasis unlike hard PL
ABRSI Hybrid PL	✓ Hard PL with PL Voting Mechanism	✓ Soft PL with $\mathcal{L}_{TE}$ and $\mathcal{L}_{DIV}$	✓ PL Voting Mechanism increases hard PL correctness ✓ Hard PL emphasises confident and correct predictions ✓ Soft PL boosts target instance participation during alignment ✓ $\mathcal{L}_{DIV}$ improves soft PL global diversity ✓ $\mathcal{L}_{TE}$ gradually increases individual soft PL certainty

its within-cluster peers, then that instance is also considered as an ambiguous near-boundary instance, and will be eliminated as shown in Fig. 4(d). Eventually, the ABRSI will only assign a hard PL to an instance if four votes reach a consensus. As shown in Fig. 4(e), the assigned hard PLs after PL voting are highly accurate. Hard PL assignment with improved accuracy can, therefore, positively guide the intrusion knowledge transfer.

Hybrid PL Strategy: Previous research efforts either used a hard-PL-only strategy that lacks the RS decision and label relationship information, or used a probabilistic soft-PL-only strategy. As summarized in Table I, the hard-PL-only strategy suffers from negative transfer caused by its erroneous assignment. Moreover, instances without hard PL agreement cannot participate in the intrusion knowledge transfer. Besides, hard PL favors easy-to-predict categories with impaired diversity. Conversely, the soft-PL-only strategy promotes better participation and diversity, but has insufficient emphasis on confident and correct predictions. If these predictions get more emphasis as in hard PL, they can positively contribute toward finer intrusion alignment.

To tackle the deficiencies summarized in Table I, the ABRSI leverages a hybrid PL strategy. Instances that have four voter agreements will be assigned with a hard PL to promote positive contributions of correct and confident predictions. The remaining instances will have a probabilistic soft PL assignment to promote full participation. The soft PL also enriches diversity, which tackles the impaired diversity caused by hard PL's potential focus on easy intrusion categories. Finally, we use a diversity maximization loss $\mathcal{L}_{DIV}$ to improve soft PL

global diversity, which is defined as follows:

$$\mathcal{L}_{DIV} = \sum_{k=1}^K p_T^{\mu(k)} \log p_T^{\mu(k)}, p_T^{\mu} = \frac{1}{n_T} \sum_{j=1}^{n_T} p_T^j \quad (7)$$

where p_T^j stands for the probabilistic output of target instance x_T^j from the shared classifier C , p_T^{μ} is the mean of all target probabilistic outputs, $p_T^{\mu(k)}$ is the k th element of p_T^{μ} . Minimizing the $\mathcal{L}_{DIV}$ promotes better global soft PL diversity.

Meanwhile, the ideal soft PL should be globally diverse while individually certain. Hence, we use α -Tsallis Entropy $\mathcal{L}_{TE}$ to promote individual soft PL certainty as follows:

$$\mathcal{L}_{TE} = \sum_{j=1}^{n_T} \left(\frac{1}{\alpha - 1} \left(1 - \sum_k p_T^{j(k)\alpha} \right) \right), \alpha > 0 \quad (8)$$

where $p_T^{j(k)}$ denotes the k th element of the probabilistic output of target instance x_T^j . The α -Tsallis entropy applies an adjustable entropic index α . A higher α value places a lower penalty on uncertain predictions, and vice versa. Hence, the ABRSI linearly decreases α between $\alpha_{\max}$ and $\alpha_{\min}$ to adaptively promote certainty for soft PLs as the training progresses.

By combining a hard PL strategy with improved accuracy, a soft PL strategy with global diversity and individual certainty, the ABRSI forms a hybrid PL strategy with improved diversity, participation, and enough emphasis on confident and correct predictions. The improved PL strategy can, therefore, benefit finer intrusion knowledge transfer.

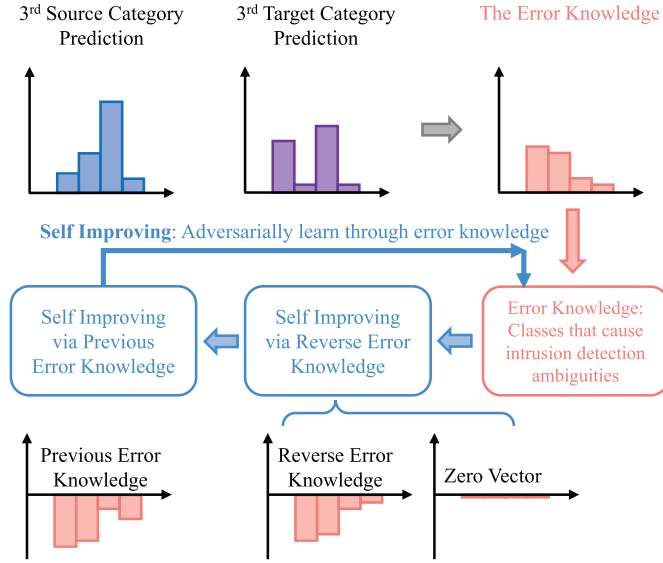


Fig. 5. Illustration of SI via EKL, the third category is chosen as an example.

EKL: Degraded intrusion knowledge transfer is stemmed from error factors that cause ambiguities. For example, as illustrated in Fig. 5, any probabilistic output other than the third category are factors that causes ambiguities and can potentially lead to compromised intrusion detection performance. To eliminate these error factors, the ABRSI model forms it as a learning process. First, the ABRSI model extracts the error knowledge by finding the categorywise divergence between probabilistic outputs between domains. Specifically, the probabilistic output of the k th source intrusion category $p_S^{\mu(k)}$ is defined as follows:

$$p_S^{\mu(k)} = \frac{1}{n_S} \sum_{i=1}^{n_S} p_S^i, \text{ where } y_S^i = k \quad (9)$$

and the probabilistic output of the k th target intrusion category $p_T^{\mu(k)}$ is defined as follows:

$$p_T^{\mu(k)} = \frac{1}{\sum_{j=1}^{n_T} p_T^j} \sum_{i=1}^{n_T} (p_T^{j(k)} \times p_T^i) \quad (10)$$

$$p_T^j = \begin{cases} \mathbb{1}(\text{PL}_{x_T^j}^{\text{NN}}), & \text{if } \text{PL}_{x_T^j}^{\text{NN}} = \text{PL}_{x_T^j}^{\text{RS}} = \text{PL}_{x_T^j}^{\text{SR}} = \text{PL}_{x_T^j}^{\text{TR}} \\ C(f(x_T^j)), & \text{otherwise} \end{cases}$$

where p_T^j is the one hot PL vector if a hard PL is assigned to x_T^j , denoted by $\mathbb{1}(\text{PL}_{x_T^j}^{\text{NN}})$, or otherwise it is the probabilistic output yielded by the shared classifier C . $p_T^{j(k)}$ denotes the k th element of p_T^j . $\text{PL}_{x_T^j}^{\text{NN}}$, $\text{PL}_{x_T^j}^{\text{RS}}$, $\text{PL}_{x_T^j}^{\text{SR}}$, $\text{PL}_{x_T^j}^{\text{TR}}$ denote the neural network predicted PL, RS decided PL, source label relationship PL and target label relationship PL, respectively. The ABRSI model then gets the category k error knowledge between probabilistic outputs, i.e., $EK^{(k)}$, defined as follows:

$$EK^{(k)} = \|p_S^{\mu(k)} - p_T^{\mu(k)}\|_2^2. \quad (11)$$

Intuitively, feeding the categorywise error knowledge vector $EK^{(k)}$ and the zero vector EK_0 into a discriminator D would

let the discriminator try to distinguish them as much as possible, i.e., exploiting the error knowledge as much as possible, and simultaneously let the feature projector E and shared classifier C to confuse the discriminator to their best extent, i.e., learn to eliminate the error knowledge. Once this minimax game reaches an equilibrium, the error knowledge can be fully exploited and then eliminated at the category level. However, we find out that merely using the zero vector EK_0 may not produce an error correction signal that is strong enough for the EKL. Therefore, we further apply a reverse error knowledge vector $EK_{\mathcal{R}}$, which stretches the divergence slightly toward the reverse direction by multiplying the error knowledge vector $EK_{\mathcal{R}}^{(k)}$ with a negative constant ψ .

We further realize that correcting the error knowledge in the current round may damage the error correction learned previously, i.e., error knowledge forgetfulness. For instance, eliminating some error factors may cause some other error factors to emerge again. Therefore, to further reinforce the error learning efficacy, we also feed the error knowledge of the previous round $\phi EK_{\mathcal{P}}^{(k)}$ to prevent error knowledge forgetfulness from happening. ϕ is a hyperparameter that balances current EKL and error knowledge forgetfulness prevention. Overall, the EKL loss is defined as follows:

$$\begin{aligned} \mathcal{L}_{\text{EKL}} &= \frac{1}{K} \sum_{k=1}^K \log(D(EK^{(k)})) \\ &\quad + \frac{1}{3K} \sum_{k=1}^K (3 - \log(D(EK_*^{(k)}))) \\ EK_*^{(k)} &\in [EK_0, EK_{\mathcal{R}}^{(k)}, EK_{\mathcal{P}}^{(k)}]. \end{aligned} \quad (12)$$

By letting the feature projector and classifier to confuse the discriminator with $EK^{(k)}$, EK_0 , $EK_{\mathcal{R}}$, and $EK_{\mathcal{P}}$ and carry out this minimax game in an adversarial manner, the EKL process forms an error knowledge extraction and elimination cycle with error knowledge forgetfulness prevention, which will gradually improve the intrusion knowledge transfer, as illustrated in Fig. 5.

C. Overall Optimization Objective

Finally, the source label can provide supervision during intrusion knowledge alignment, which is defined as follows:

$$\mathcal{L}_{\text{SUP}} = \frac{1}{n_S} \sum_{i=1}^{n_S} \mathcal{L}_{\text{CE}}(C(f(x_S^i)), y_S^i) \quad (13)$$

where $\mathcal{L}_{\text{CE}}$ denotes cross-entropy loss. The overall optimization objective of the ABRSI model is as follows:

$$\begin{aligned} \min_{E_S, E_T, C} & (\mathcal{L}_{\text{SUP}} + \rho \mathcal{L}_{\text{ABR}} + \delta \mathcal{L}_{\text{DIV}} + \tau \mathcal{L}_{\text{TE}} + \gamma \mathcal{L}_{\text{EKL}}) \\ \max_D & (\gamma \mathcal{L}_{\text{EKL}}) \end{aligned} \quad (14)$$

where ρ , γ , δ , and τ are hyperparameters controlling the weight of the corresponding components. To achieve an end-to-end optimization process, a gradient reversal layer [29] is added for the discriminator, which performs the role of an identity function during forward propagation, and reverses the gradient during backpropagation. Once the above minimax

game reaches an equilibrium, the intrusion knowledge can be transferred in a fine-grained manner, so that the shared classifier can perform intrusion detection as accurate as possible.

V. EXPERIMENT

We utilize five comprehensive and representative intrusion detection data sets and nine state-of-the-art comparing baseline methods to verify the effectiveness of the ABRSI model. In addition, the contribution and necessity of each ABRSI constituting component, and the computational efficiency of the ABRSI model is verified.

A. Experimental Data Sets

We use five comprehensive intrusion detection data sets, including three NI data sets: 1) NSL-KDD; 2) UNSW-NB15; and 3) CICIDS2017, and two IID data sets: 1) UNSW-BOTIOT and 2) UNSW-TONIOT.

NI Data Set (NSL-KDD): Released in 2009, the NSL-KDD data set [30] has a better data quality than its previous version [31]. The data set contains benign traffic with four types of representative intrusions, such as denial-of-service attack, probing attack, etc. We utilize 20% of the data set, a reasonable amount following other works such as [32]. The data set adopts a 41-D feature representation. We follow Harb [33] to use the top-31 most informative features as the feature space. The data set is denoted as K .

NI Data Set (UNSW-NB15): The UNSW released this data set [34] in 2015, captured using the Ixia PerfectStorm tool, a comprehensive security test platform commonly used by the industry. The data set has high quality and up-to-date comprehensive network flows. It contains normal network traffic and nine categories of representative intrusions, such as DoS attacks, reconnaissance attacks, etc. Following the data set magnitude in [35], we utilize 6500 traffic records. We also perform preprocessing to reduce the feature dimension from 49 to 45 by removing four features having a value zero for nearly all records. The data set is denoted as N .

NI Data Set (CICIDS2017): As one of the most up-to-date NI data sets, this data set [36] was released in 2017. The data set captures seven intrusion types, represented in 77 dimensions. We utilize 20% of the data set provided by its creator. The preprocessing steps including data deduplication and categorical-numerical data conversion. Guided by Stiawan [37], we use features with the top 40 information gain as the feature representation. The data set is denoted as C .

II Data Set (UNSW-BOTIOT): This data set [38] was released in 2017. It contains up-to-date modern attack models, tested on a realistic testbed that contains commonly used IoT devices, such as weather station, smart fridge, etc. The testbed also uses the MQTT protocol, a lightweight communication protocol commonly used by IoT. The data set has four categories of modern intrusion types, including information theft attacks, DoS attacks, etc. We follow [35] to use around 10 000 data records. We follow data set creator's advice to use the top ten most informative features among 46 original features and denote the data set as B .

II Data Set (UNSW-TONIOT): The UNSW-TONIOT [39] data set is one of the most up-to-date II data sets, released in 2021. It involves the latest IoT standards, protocols and uses a sophisticated testbed, consisting of seven types of real IoT devices, including modbus sensors, weather meters, GPS trackers, etc. The data set contains nine types of common IIs [40], such as scanning attacks, DoS attacks, etc. Each type of IoT device captures data in its own dimension, causing heterogeneities between data domains. Following [41], we utilize around 10% of the data set, and select three IoT devices, i.e., weather meter, GPS tracker and modbus sensor, denoted as W , G and M , respectively.

Data Set Comprehensiveness and Intrusion Methods: The data sets we used are comprehensive and representative, making them sufficient to verify the proposed model. First, these data sets are widely used and recognized by the intrusion detection research community with thousands of citations. Second, all data sets are developed and released in recent years, some of them are released in 2021. Hence, these data sets reflect modern attack methods. Third, these data sets are captured using widely recognized testbeds with comprehensiveness. The IoT data sets also involve real-world IoT devices, deployed in realistic environments. Finally, the network and IoT data sets have at most eight shared intrusion categories transferrable as intrusion knowledge. These intrusions are generated using various different mechanisms to reflect the diversity of modern intrusions, and have a wide coverage of modern intrusion trends, i.e., with a 100%, 55%, 100%, 100% and 98% coverage on NSL-KDD, UNSW-NB15, CICIDS2017, UNSW-BOTIOT, and UNSW-TONIOT data sets, respectively. Together, these rationales make these data sets sufficient to testify the effectiveness of the proposed method.

B. Implementation Details

We use the deep learning framework PyTorch to implement the ABRSI model. We follow [10], [25] to implement the feature projectors as two-layer neural networks with LeakyRelu activation function, and implement both the shared classifier C and EKL discriminator D as single-layer neural networks. All experiments use a single set of hyperparameters, which are as follows: $\rho_{\max} = 0.1$, $\delta = 1$, $\tau = 0.005$, $\gamma = 0.1$, $\text{TopN} = 3$, $\#\text{PL}^{\text{SR}} \text{ neighbor} = 3$, $\alpha_{\max} = 8$, $\alpha_{\min} = 4$, $\psi = -0.3$, and $\phi = -0.05$. Note that the hyperparameter ρ will linearly increase from 0 to $\rho_{\max}$ to gradually emphasize the importance of ABR matching as the recommendation becomes more mature as training progresses. Besides, hyperparameter α will linearly decrease from $\alpha_{\max}$ to $\alpha_{\min}$ to gradually increase the soft PL individual certainty as the training evolves. We also perform parameter sensitivity analysis to verify the stability and robustness of ABRSI on hyperparameter settings. During our experiments, we consider both binary intrusion detection, i.e., distinguish benign traffic and intrusion traffic as a whole, and multiclass intrusion detection, i.e., distinguish benign traffic and each individual intrusion type. We denote multiclass intrusion detection with superscript (*) on the target domain. Following [10], [35], and [42], we utilize unlabeled target

TABLE II
INTRUSION DETECTION ACCURACY OF TEN METHODS ON NINE TASKS

$\mathcal{D}_S \rightarrow \mathcal{D}_T$	$N \rightarrow B^*$	$N \rightarrow G$	$N \rightarrow W$	$N \rightarrow M$	$C \rightarrow G$	$C \rightarrow B$	$K \rightarrow B^*$	$K \rightarrow G^*$	$K \rightarrow W$	Avg
CDAN	40.17	49.89	52.90	50.23	49.75	49.65	34.22	24.84	53.78	45.05
MCC	35.96	50.02	50.18	51.20	49.85	57.88	39.66	24.86	53.55	45.61
CAFD	33.60	49.72	55.51	49.90	50.10	60.34	36.38	24.80	50.30	45.63
JADA	32.20	50.10	51.40	51.55	50.80	57.85	38.52	24.80	56.04	45.92
CAFA	36.45	49.15	53.45	51.28	50.88	62.34	38.32	24.85	50.43	46.35
DCAN	34.97	51.00	49.32	51.25	50.30	59.38	40.20	24.90	55.85	46.35
GDCAN	37.23	49.75	51.03	51.68	50.20	60.26	38.43	24.85	56.50	46.66
PDA	40.06	71.88	59.38	52.43	60.66	54.44	43.84	31.25	55.79	52.19
ADAR	45.75	70.66	55.92	51.54	63.21	58.00	42.04	39.73	57.40	53.81
ABRSI (Ours)	46.93	90.31	67.45	53.17	88.04	65.42	47.42	47.27	61.22	63.03

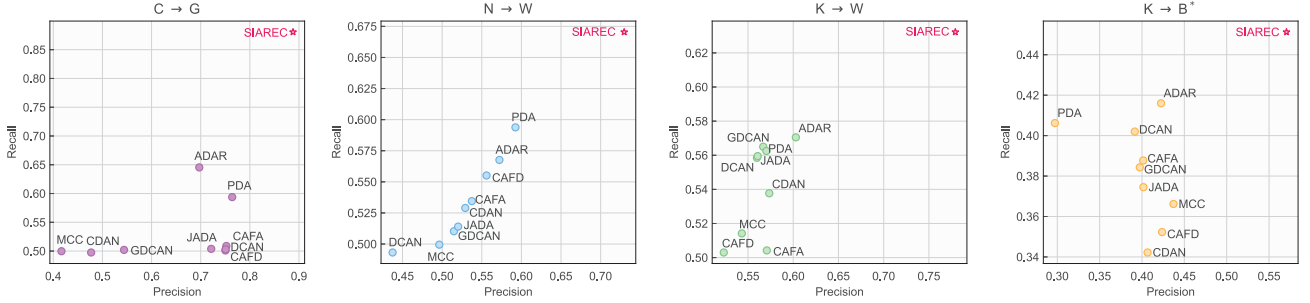


Fig. 6. Intrusion detection performance evaluation using precision (x-axis) and recall (y-axis) of ten methods on four randomly selected tasks.

domain prediction accuracy, category-weighted precision (P), recall (R), F1-Score (F), and Area under the ROC Curve (A) as the evaluation metrics.

C. State-of-the-Art Baselines

We use nine state-of-the-art baselines to verify the superiority of the ABRSI model, which include CDAN [15], MCC [16], CAFA [13], CAFD [13], JADA [17], DCAN [20], GDCAN [21], PDA [14], and ADAR [24]. All methods are from top-tier venues, and eight of them are published between 2019 and 2022. We summarize their differences with ABRSI as follows.

- 1) From the RS usage perspective, the ADAR utilizes domain discriminator to facilitate the learning of a RS, and lets it perform intrusion category recommendation. However, the ADAR uses a single RS and lacks ABR matching. It also fails to use RS to facilitate better PL prediction.
- 2) From the PL perspective, methods such as CDAN utilize pure hard PL, while methods, such as MCC and PDA utilize pure soft PL. None of them consider the potential of a hybrid PL strategy. Hard PL methods also assign their PLs based on direct network prediction, which fail to consider the label relationship information, resulting in error-prone PLs. Besides, due to the ignorance of RS, none of these methods utilize RS to benefit PL assignment.
- 3) From the feature alignment perspective, methods, such as JADA, CAFA, CAFD, DCAN, and GDCAN minimize probabilistic divergence via Wasserstein distance

minimization, direct divergence loss minimization, etc. Methods, such as ADAR, PDA, JADA, CAFA, CAFD, CDAN, etc., focus on domain discrimination-based feature alignment. However, using adversarial learning to facilitate better feature alignment via EKL, and simultaneously avoid error knowledge forgetfulness is a void that needs to be filled.

Therefore, these state-of-the-art baselines are representative and sufficient to verify the ABRSI method's effectiveness.

D. Intrusion Detection Performance

We first present the intrusion detection accuracy results on nine randomly selected tasks in Table II. As we can observe, the ABRSI model significantly outperforms all state-of-the-art counterparts. Specifically, it outperforms the best comparing method ADAR by 9.2%, demonstrating the best intrusion detection effectiveness. It's natural to observe this since the ADAR method fails to use the ABR matching mechanism, and ignores the usefulness of the RS-based PL assignment strategy. It also lacks sufficient EKL to promote finer intrusion knowledge transfer, resulting in compromised intrusion detection performance.

We then present the performance measured using precision and recall in Fig. 6, in which the x-axis and y-axis indicate precision and recall, respectively. As we can see, the ABRSI method sits at the top-right corner, demonstrating the best precision and recall compared with other methods with a significant performance improvement. The best precision performance indicates the most intrusions flagged by the ABRSI are correct decisions, while the best recall performance

TABLE III
INTRUSION DETECTION PERFORMANCE EVALUATION USING F1-SCORE
OF TEN METHODS ON FOUR RANDOMLY SELECTED TASKS

$\mathcal{D}_S \rightarrow \mathcal{D}_T$	C $\rightarrow$ G	N $\rightarrow$ W	K $\rightarrow$ W	K $\rightarrow$ B*	Avg
CDAN	0.49	0.53	0.56	0.37	0.49
MCC	0.45	0.50	0.53	0.40	0.47
CAFD	0.60	0.56	0.51	0.39	0.50
JADA	0.59	0.52	0.56	0.39	0.52
CAFA	0.61	0.54	0.54	0.39	0.52
DCAN	0.60	0.46	0.56	0.40	0.51
GDCAN	0.52	0.51	0.57	0.39	0.50
PDA	0.67	0.59	0.57	0.34	0.54
ADAR	0.62	0.57	0.59	0.42	0.55
ABRSI	0.88	0.70	0.70	0.51	0.70

TABLE IV
INTRUSION DETECTION PERFORMANCE EVALUATION USING AUC SCORE
OF NINE METHODS ON FOUR RANDOMLY SELECTED TASKS

$\mathcal{D}_S \rightarrow \mathcal{D}_T$	C $\rightarrow$ G	N $\rightarrow$ W	K $\rightarrow$ W	K $\rightarrow$ B*	Avg
CDAN	0.47	0.51	0.53	0.50	0.50
MCC	0.51	0.48	0.46	0.54	0.50
CAFD	0.63	0.56	0.43	0.50	0.53
JADA	0.50	0.55	0.54	0.54	0.53
CAFA	0.62	0.55	0.57	0.55	0.57
DCAN	0.51	0.44	0.55	0.58	0.52
GDCAN	0.50	0.49	0.58	0.56	0.53
PDA	0.42	0.55	0.52	0.59	0.52
ABRSI	0.86	0.70	0.59	0.65	0.70

means that the ABR SI model can flag as many intrusions as possible.

We also calculate the F1-score, a harmonic mean that reflects the balance between precision and recall, and present the result in Table III. The ABR SI method has the best F1-score performance, outperforming the best counterpart PDA and ADAR by 29.6% and 27.3%, respectively. The best F1-score indicates that the ABR SI method can balance intrusion flagging and false-alarm avoidance properly, demonstrating its real-world usefulness.

Finally, we measure the AUC score, an evaluation metric reflecting the intrusion detection capability. The results are presented in Table IV. Note that the ADAR yields ranked recommendation, not a probabilistic prediction, hence, it is not applicable to AUC measurement. The ABR SI shows superior performance compared with all counterparts, yielding a 34.6% AUC improvement, demonstrating the best ability to detect intrusions with an excellent separability.

Overall speaking, the superior performances of ABR SI on all evaluation metrics indicate that the ABR SI model can detect intrusions as accurate as possible. It flags as many intrusions as possible from all possible malicious behaviors while avoiding triggering too much false alarms. It can balance intrusion flagging and false-alarm avoidance properly, thanks to its best ability of intrusion separability. Therefore, the results demonstrate the real-world applicability of the ABR SI model for effective IID.

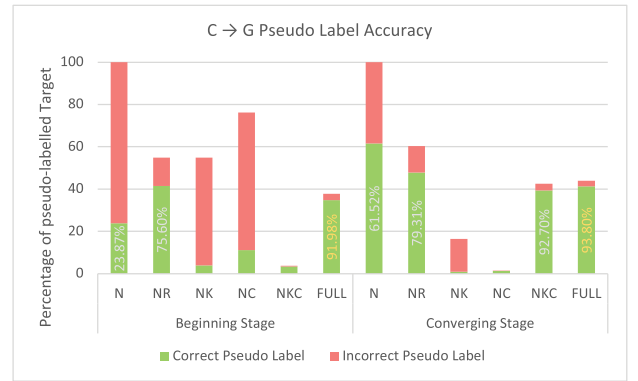


Fig. 7. PL accuracy visualization. The bar height indicates ratio of hard PL assigned target instances, green and red color indicate correctly and incorrectly assigned PL ratio, respectively.

E. Adaptive Bi-Recommendation Effectiveness

We now analyze the effectiveness of the ABR mechanism, and the contribution it makes on improving the quality of hard PL assignment. The results are indicated in Table V. The Full ABR SI outperforms all the ABR-ablated counterparts by a large margin. Specifically, the ABR and RS-based hard PL assignment contribute 3.8% and 4% of performance improvement. Besides, completely lacking the ABR mechanism degrades the performance by 5%. It, therefore, verifies the positive contribution made by the ABR mechanism, and the usefulness of its constituting components.

F. Self-Improving: Pseudo Label Voting Mechanism Effectiveness

Voting Component Effectiveness: We verify the effectiveness of each hard PL voting component, and present the results in Table VI. As we can observe, the full ABR SI shows superior performance than its voting-ablated counterparts by a large margin, demonstrating that all voters are indispensable for an accurate hard PL assignment outcome.

To further illustrate the hard PL accuracy, we present the hard PL accuracy, hard PL ratio and overall intrusion detection accuracy during the beginning and converging stages in Table VII, and visualize the result of a randomly selected task in Fig. 7. As we can observe, the full ABR SI can yield accurate hard PL assignment even at the beginning training stage. It will not blindly increase the percentage of hard PL-assigned instances, since the accuracy of hard PL matters more than the pure amount. As the training progresses, the hard PL accuracy of full ABR SI can even reach 99%, the highest among all its ablated counterparts. Besides, the hard PL ratio is also gradually increased. By assigning hard PL in an accurate manner, the full ABR SI leads to the best intrusion detection performance, thanks to the contribution made by each hard PL voting component.

Hard PL Diversity: We also find out that the ABR SI can yield hard PL assignment with a higher diversity than the hard-PL-only assignment without voting. We use the Hellinger distance to measure the diversity, i.e., the distance between the hard PL distribution and the ideal distribution of target II

TABLE V
ABLATION GROUP A: ABR EFFECTIVENESS

Experiment	Adaptive Bi-Recommendation Usage		Ablation Results				Avg
	ABR Matching	RS-based PL ^{RS}	N → W	C → G	C → B	K → B*	
A ₁	×	✓ (PL ^{NN} +PL ^{RS} +PL ^{SR} +PL ^{TR})	63.08	87.27	62.22	40.60	63.29
A ₂	✓	×	62.63	87.47	60.34	41.83	63.07
A ₃	×	×	59.56	87.13	61.56	40.23	62.12
ABRSI Full	✓	✓	67.45	88.04	65.42	47.42	67.08

TABLE VI
ABLATION GROUP B: PSEUDO LABELING VOTING COMPONENT EFFECTIVENESS

Experiment	PL Voting Components				Ablation Results				Avg
	PL ^{NN}	PL ^{RS}	PL ^{SR}	PL ^{TR}	N → W	C → G	C → B	K → B*	
B ₁	✓	✓	×	×	59.61	86.91	63.11	41.38	62.75
B ₂	✓	×	✓	×	56.02	72.58	55.59	40.15	56.09
B ₃	✓	×	×	✓	56.10	87.06	59.32	38.08	60.14
ABRSI Full	✓	✓	✓	✓	67.45	88.04	65.42	47.42	67.08

TABLE VII
HARD PL ACCURACY, RATIO, AND OVERALL PREDICTION ACCURACY UNDER DIFFERENT PL VOTING STRATEGIES

$\mathcal{D}_S \rightarrow \mathcal{D}_T$	PL Strategy	Beginning Stage			Converging Stage		
		Hard PL Acc	Hard PL Ratio	Overall Acc	Hard PL Acc	Hard PL Ratio	Overall Acc
N → W	PL ^{NN}	46.42	100.00	46.42	51.00	100.00	51.00
	PL ^{NN} Soft	0.00	0.00	51.10	0.00	0.00	54.45
	PL ^{NN} +PL ^{RS}	79.31	3.62	52.12	58.40	62.08	61.21
	PL ^{NN} +PL ^{SR}	22.93	15.86	50.10	53.71	67.64	55.03
	PL ^{NN} +PL ^{TR}	47.02	89.56	46.79	55.09	27.02	56.18
	PL ^{NN} +PL ^{SR} +PL ^{TR}	51.82	64.61	54.65	63.84	45.27	63.73
	FULL	76.62	8.13	52.76	98.92	14.44	71.02
C → G	PL ^{NN}	23.87	100.00	23.87	61.52	100.00	61.52
	PL ^{NN} Soft	0.00	0.00	41.24	0.00	0.00	80.42
	PL ^{NN} +PL ^{RS}	75.60	54.87	78.10	79.31	60.36	85.38
	PL ^{NN} +PL ^{SR}	9.60	40.64	53.90	5.80	16.39	72.49
	PL ^{NN} +PL ^{TR}	14.68	76.17	32.83	91.07	1.40	86.48
	PL ^{NN} +PL ^{SR} +PL ^{TR}	90.55	3.85	53.20	92.70	42.54	87.18
	FULL	91.98	37.76	77.90	93.80	43.99	88.54

TABLE VIII
HARD PL DIVERSITY MEASURED USING THE HELLINGER DISTANCE

Experiment	N → W	C → G	C → B	K → B*	Avg
Hard PL ^{NN}	0.26	0.20	0.38	0.57	0.35
ABRSI Full	0.20	0.15	0.08	0.47	0.23

domain. The smaller the Hellinger distance is, the better the diversity. The Hellinger distance has the following definition:

$$H(P_{PL}, P_T) = \frac{1}{\sqrt{2}} \sqrt{\sum_{k=1}^K \left(\sqrt{P_{PL}^{(k)}} - \sqrt{P_T^{(k)}} \right)^2} \quad (15)$$

where P_{PL} and P_T denote the hard PL distribution and target II intrusion distribution, $P_{PL}^{(k)}$ denotes the k th element of P_{PL} . The results are shown in Table VIII. As we can observe, the full ABRSI yields an improved hard PL diversity compared with its hard-PL-only-based counterpart by reducing the Hellinger Distance by 34.3%. The reason is that the hard-PL-only-based counterpart is more likely to focus on easy-to-predict categories without the refinement achieved by the voting mechanism. Conversely, the full ABRSI is assisted by the hard PL voting mechanism, which can improve the hard

PL accuracy and meanwhile reduce the over-emphasis on easy-to-predict categories, which leads to better hard PL diversity, and, hence, benefits the intrusion detection effectiveness.

G. Self-Improving: Hybrid Pseudo Label Effectiveness

We now verify the superiority of the ABRSI model compared with its hard-PL-only and soft-PL-only counterpart and present the results in Table IX. As we can notice, the full ABRSI model has the best intrusion detection accuracy, benefitted by the hybrid PL strategy. Its hard-PL-only and soft-PL-only peer show a 16.9% and 9.7% accuracy drop, demonstrating that both components constituting the hybrid PL strategy are indispensable.

H. Self-Improving: Soft PL Diversity and Confidence

We now verify the effectiveness of soft PL diversity and confidence improvement, part of the SI mechanism. The intrusion detection accuracy results are presented in Table X. The diversity and confidence improvement contribute 11.6% and 4.1% of accuracy improvement, respectively. Without both improvements, the intrusion detection accuracy will even drop by 12%.

TABLE IX
ABLATION GROUP C: HYBRID PSEUDO LABELING EFFECTIVENESS

Experiment	PL Types		Ablation Results				Avg
	Involving Hard PL	Involving Soft PL	$N \rightarrow W$	$C \rightarrow G$	$C \rightarrow B$	$K \rightarrow B^*$	
C_1	✓	✗	50.92	61.52	52.39	35.86	50.17
C_2	✗	✓	53.74	80.42	55.37	40.05	57.40
ABRSI Full	✓	✓	67.45	88.04	65.42	47.42	67.08

TABLE X
ABLATION GROUP D: SOFT PL CONFIDENCE AND DIVERSITY SI COMPONENTS

Experiment	Soft PL confidence and diversity improvement component		Ablation Results				Avg
	Confidence Improvement ($\mathcal{L}_{TE}$)	Diversity Improvement ($\mathcal{L}_{DIV}$)	$N \rightarrow W$	$C \rightarrow G$	$C \rightarrow B$	$K \rightarrow B^*$	
D_1	✗	✓	58.88	87.80	64.00	41.21	62.97
D_2	✓	✗	55.77	69.47	57.47	39.06	55.44
D_3	✗	✗	54.95	76.83	53.09	35.64	55.13
ABRSI Full	✓	✓	67.45	88.04	65.42	47.42	67.08

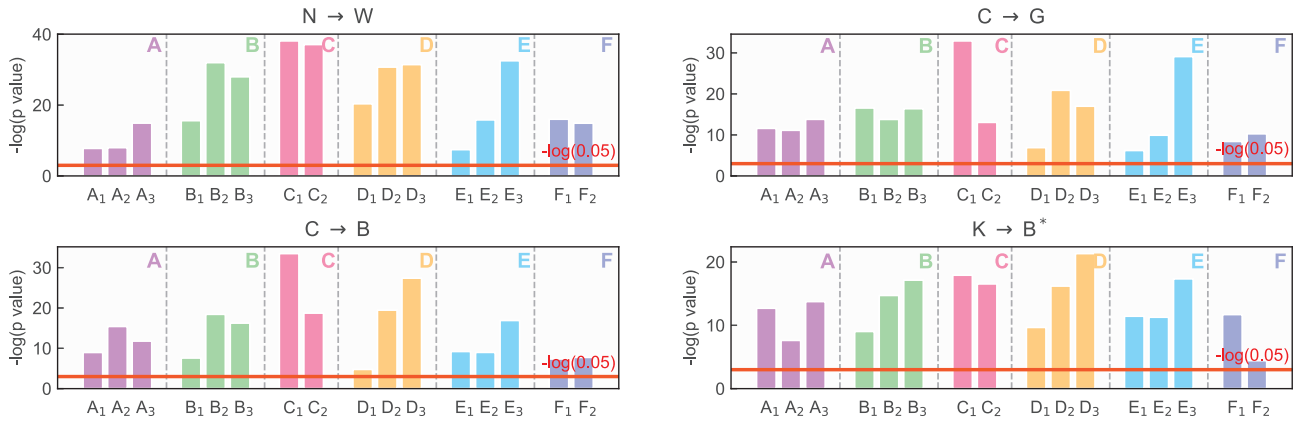


Fig. 8. Hypothesis testing for ablation groups A–F. The red line denotes the significance threshold $-\log(0.05)$.

TABLE XI
SOFT PL DIVERSITY MEASURED USING THE HELLINGER DISTANCE

Experiment	$N \rightarrow W$	$C \rightarrow G$	$C \rightarrow B$	$K \rightarrow B^*$	Avg
$ABRSI-\mathcal{L}_{TE}$	0.27	0.32	0.41	0.43	0.36
ABRSI Full	0.13	0.04	0.19	0.36	0.18

TABLE XII
SOFT PL CERTAINTY MEASURED USING PERCENTAGE OF HIGH CERTAINTY PREDICTIONS (PROBABILITY THRESHOLD = 70%)

Experiment	$N \rightarrow W$	$C \rightarrow G$	$C \rightarrow B$	$K \rightarrow B^*$	Avg
$ABRSI-\mathcal{L}_{DIV}$	0.21	0.00	0.81	0.01	0.26
ABRSI Full	0.25	0.05	0.97	0.06	0.33

Besides, the Hellinger Distance is also used to measure the diversity of soft PL with results presented in Table XI, and the confidence is measured using the percentage of high certainty predictions with certainty threshold set to 0.7 and the results are given in Table XII. As we can observe, the diversity improvement component decreases the Hellinger Distance by 50%, and the confidence improvement component achieves higher prediction certainty, with 36% of the intrusion detection decision having a certainty greater than 70%. Hence, these SI

components achieve a globally diverse and individually certain soft PL assignment, which boosts the intrusion detection performance.

I. Error Knowledge Learning Necessity and Effectiveness

EKL Necessity: We verify the necessity of the EKL by comparing it with three counterparts: E_1 , which ablates the EKL; E_2 , which uses a domain discriminator that distinguishes the domain origin of instances; E_3 , which directly minimizes the Euclidean distance between probabilistic outputs. The results are illustrated in Table XIII. As we can see, the ABRSI achieves the best intrusion detection accuracy with a 6.7% accuracy improvement on average. It is natural to observe this since both E_1 and E_2 lack consideration of error knowledge mining and elimination, while E_3 eliminates error knowledge in a coarse-grained manner. It in turn verifies the necessity of the EKL in the ABRSI model.

EKL Constituting Component Effectiveness: We verify the necessity of EKL assisted by previous error knowledge EK_P and reverse error knowledge EK_R . As shown in Table XIV, the EK_P and EK_R contribute 4.5% and 4% intrusion accuracy improvement. Hence, it verifies that the reverse error knowledge can effectively strengthen the error knowledge

TABLE XIII
ABLATION GROUP E: EKL NECESSITY ANALYSIS

Experiment	Involve Discriminator	Description	Ablation Results				Avg
			$N \rightarrow W$	$C \rightarrow G$	$C \rightarrow B$	$K \rightarrow B^*$	
$E_1: \gamma = 0$	✓	Without EKL	64.21	87.76	62.18	40.52	63.67
E_2 : Domain Dis	✓	With domain discriminator	60.47	87.50	62.20	41.14	62.83
E_3 : Prob Matching	✗	With probabilistic output matching	54.74	65.32	58.35	39.65	54.52
ABRSI Full	✓	With EKL	67.45	88.04	65.42	47.42	67.08

TABLE XIV
ABLATION GROUP F: EKL CONSTITUTING COMPONENT EFFECTIVENESS ANALYSIS

Experiment	EKL		Ablation Results				Avg
	EK_P	EK_R	$N \rightarrow W$	$C \rightarrow G$	$C \rightarrow B$	$K \rightarrow B^*$	
F_1	✗	✓	59.31	87.45	62.84	40.77	62.59
F_2	✓	✗	60.72	87.43	62.34	42.00	63.12
ABRSI Full	✓	✓	67.45	88.04	65.42	47.42	67.08

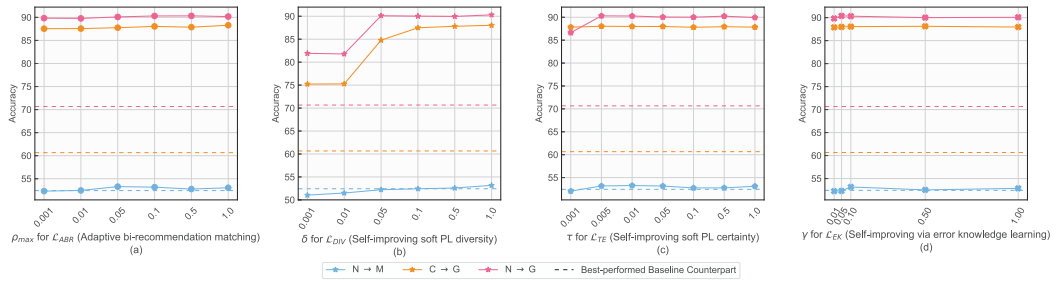


Fig. 9. Parameter sensitivity verification for balancing hyperparameters ρ , δ , τ , and γ in the overall optimization objective.

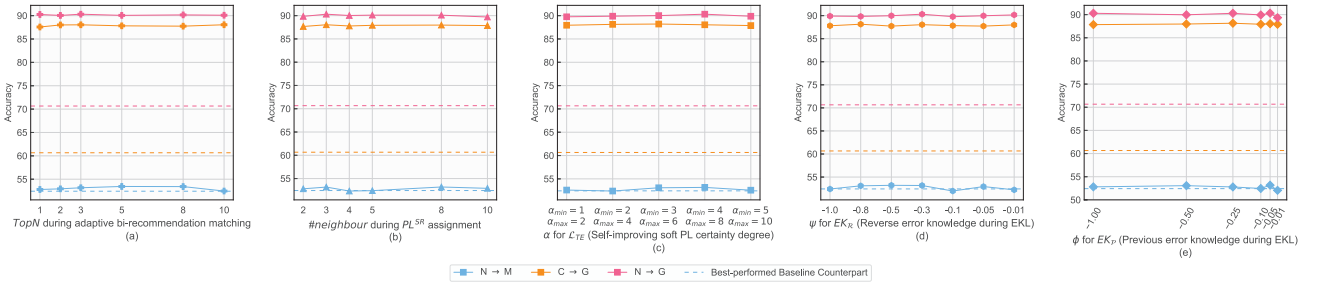


Fig. 10. Parameter sensitivity verification for other hyperparameters.

TABLE XV
COMPUTATIONAL EFFICIENCY: TRAINING TIME PER EPOCH (SECOND)

$\mathcal{D}_S \rightarrow \mathcal{D}_T$	$N \rightarrow B^*$	$N \rightarrow G$	$N \rightarrow M$	$C \rightarrow G$	$K \rightarrow B^*$	$K \rightarrow W$	Avg
PDA	1.67	1.73	1.67	1.72	1.74	1.67	1.7
ADAR	0.17	0.10	0.07	0.07	0.17	0.09	0.11
ABRSI	0.36	0.19	0.21	0.20	0.35	0.21	0.25

elimination, while the error knowledge forgetfulness is effectively prevented by the previous EKL, leading to more accurate intrusion detection.

J. Hypothesis Testing of Contributions

We perform significance T-test with the significance threshold of 0.05 to verify the statistical significance of the contributions made by each ABRSI constituting component. The results are illustrated in Fig. 8. The red horizontal line indicates the significance threshold $-\log(0.05)$, the higher the bar is beyond the red horizontal line, the more significant is the

contribution made by that ablated component. We can observe that all bars are significantly higher than the red line, demonstrating statistical soundness of the contributions made by each constituting component of the ABRSI.

K. Hyperparameter Sensitivity Analysis

We also verify the stability and robustness of the ABRSI model toward hyperparameter settings. The parameter sensitivity results of loss balancing hyperparameters are illustrated in Fig. 9, and the parameter sensitivity of all other hyperparameters are presented in Fig. 10. The colored dashed lines indicate

TABLE XVI
COMPUTATIONAL EFFICIENCY: INFERENCE TIME PER INSTANCE (MILLISECOND = 10^{-3} SECOND)

$\mathcal{D}_S \rightarrow \mathcal{D}_T$	$N \rightarrow B^*$	$N \rightarrow G$	$N \rightarrow M$	$C \rightarrow G$	$K \rightarrow B^*$	$K \rightarrow W$	Avg
PDA	1.12	1.09	1.25	1.09	1.13	1.06	1.12
ADAR	2.69×10^{-2}	1.75×10^{-2}	1.68×10^{-2}	1.72×10^{-2}	2.61×10^{-2}	1.72×10^{-2}	2.03×10^{-2}
ABRSI	2.56×10^{-4}	1.77×10^{-4}	2.68×10^{-4}	1.83×10^{-4}	2.08×10^{-4}	2.30×10^{-4}	2.20×10^{-4}

TABLE XVII
ACRONYM TABLE

Acronym	Interpretation
ABRSI	Adaptive Bi-Recommendation and Self-Improving Network
HDA	Heterogeneous Domain Adaptation
IID	IoT Intrusion Detection
NID	Network Intrusion Detection
DA	Domain Adaptation
NI	Network Intrusion
II	IoT Intrusion
RS(s)	Recommender System(s)
PL(s)	Pseudo Label(s)
ABR	Adaptive Bi-Recommendation Matching Mechanism
SI	Self-Improving Mechanism
EK	Error Knowledge
EKL	Error Knowledge Learning
NN	Neural Network
SR	Source label Relationship
TR	Target label Relationship

TABLE XVIII
NOTATION TABLE

Notation	Interpretation
$\mathcal{D}_*$	Domain, $*$ $\in \{S, T\}$
x_{S_i}	i^{th} instance in source domain
y_{S_i}	The intrusion category label of the i^{th} source instance
n_S	Number of source domain instances
d_S	Dimension of source domain
$E_S(x_i)$	The source projector
$f(x_i)$	Features projected by the projector
d_C	Dimension of common feature space
RS_S	Source-trained recommender system
M_S	Original feature matrix of source domain
R	Reduced dimension parameter during LSI
U^R	The feature-latent matrix with reduced dimension R
T^R	The latent transfer matrix with reduced dimension R
$V^{R\top}$	The instance-latent matrix with reduced dimension R
$x_S^{i'}$	Transformed source instance x_S^i by the LSI algorithm
$RS_S(x_T^j)$	The recommendation for x_T^j provided by source-trained RS_S
$PL_{x_T^j}^{RS}$	The RS-based PL for x_T^j
$x_S^{(i)}$	The category centroid for the i^{th} source category
$RS_T(x_S^{(i)})^n$	The top N recommendation for source centroid $x_S^{(i)}$ provided by target-trained RS_T
$\mathcal{L}_{ABR}$	The adaptive bi-recommendation matching loss
P_T^j	The probabilistic output of x_T^j from the classifier C
P_T^μ	The mean of all target probabilistic outputs
$p_T^{\mu(k)}$	The k^{th} element of p_T^μ
$\mathcal{L}_{DIV}$	The diversity maximisation loss
$\mathcal{L}_{TE}$	The Tsallis Entropy loss
α	The entropic index in $\mathcal{L}_{TE}$
$\mathbb{1}()$	One hot vector
PL^{NN}	The neural network-predicted PL
PL^{RS}	The recommender system-decided PL
PL^{SR}	The source label relationship information-based PL
PL^{TR}	The target label relationship information-based PL
$EK^{(k)}$	The error knowledge of intrusion category k
EK_0	The zero error knowledge vector
$EK_{\mathcal{R}}^{(k)}$	The reverse error knowledge of intrusion category k
$EK_{\mathcal{P}}^{(k)}$	The previous error knowledge of intrusion category k
ψ	The negative constant applied for $EK_{\mathcal{R}}^{(k)}$
ϕ	The negative constant applied for $EK_{\mathcal{P}}^{(k)}$
$\mathcal{L}_{EKL}$	The error knowledge learning loss
$\mathcal{L}_{SUP}$	The source domain supervision loss
$\mathcal{L}_{CE}$	Cross entropy loss
$\rho, \gamma, \delta, \tau$	Balancing hyperparameter for $\mathcal{L}_{ABR}$, $\mathcal{L}_{EK}$, $\mathcal{L}_{DIV}$ and $\mathcal{L}_{TE}$, respectively

the corresponding best performed baseline counterpart for each task. As we can observe, the ABRSI presents very stable performance when hyperparameters vary within their reasonable ranges. Besides, the ABRSI stably outperforms its best baseline counterpart, hence, it verifies the ABRSI performs stably under varied hyperparameter settings.

Besides, when tackling different data domains that present heterogeneities, all experiments share a fixed set of hyperparameter setting, and the ABRSI model is still capable to achieve satisfying results without the need to constantly reset hyperparameters. Hence, it further demonstrates the robustness of ABRSI in terms of hyperparameter setting.

L. Intrusion Detection Efficiency

Finally, we evaluate the computational efficiency of the ABRSI model from two aspects, i.e., the training time taken by each training epoch, and the inference time used when performing intrusion detection on each instance. The results are presented in Tables XV and XVI, respectively. Note that only the top-2 best performed baselines are compared for computational efficiency. As we can observe, although the ABRSI model is not the fastest in terms of training speed, its training speed is not significantly slow. Besides, the training can be done on devices with strong computational capability. Therefore, the training efficiency of ABRSI is acceptable. On the other hand, the ABRSI achieves the fastest intrusion detection speed, which demonstrates its usefulness when performing intrusion detection for IoT scenarios. Overall, the ABRSI only involves a neural network structure that is relatively shallow, the RS it utilized is not neural network-based

and has a low computational complexity compared with other NN-based RS counterparts. Therefore, the satisfying computational efficiency of the ABRSI model makes its deployment to devices such as network gateways feasible, and, hence, is practical for real-world IID.

VI. CONCLUSION

In this article, we form the IID problem as an unsupervised HDA problem that transfers enriched intrusion knowledge from a NI domain to facilitate better IID under data-scarcity. First, an ABR matching mechanism matches bi-recommendation interests and in turn enforces intrusion feature alignment. They can mutually benefit each other, forming a positive loop that can facilitate finer intrusion knowledge transfer. A SI mechanism with four components is also utilized. Specifically, a hard PL voting mechanism aims to improve the accuracy and diversity of hard PL assignment. A hybrid PL strategy introduces better participation and diversity, while preserving enough emphasis on confident and correct predictions. The diversity and certainty of soft PL is also jointly improved to achieve globally diverse and individually certain soft PL. Finally, an EKL mechanism is used to exploit factors that cause prediction ambiguities, and adaptively conduct error knowledge exploitation, elimination and error knowledge forgetfulness prevention. Holistically, these components form the ABRSI model that can boost intrusion detection performance under the IoT scenarios. Comprehensive experiments on five widely adopted intrusion detection data sets are conducted and compared against nine state-of-the-art baselines to verify the effectiveness of the ABRSI model. The contribution of each ABRSI constituting component is also verified with statistical significance. Finally, the computational efficiency is examined to show the ABRSI's applicability.

APPENDIX

Acronym Table and Notation Table: We provide an acronym table and a notation table for better readability.

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